

Chapter 128 — Special Urologic Considerations in Transmasculine Individuals

- **Gender identity** is a person's own sense of gender, which may not conform to the sex assigned at birth, its gender role, or primary and secondary sex characteristics.
- **Sex versus gender:** sex denotes biological differences (chromosomes, hormonal profile, internal and external sex organs); gender denotes the characteristics a society or culture designates as masculine or feminine.
- **Gender dysphoria (GD)** is the mismatch between gender identity and sex; significant distress meets criteria for a formal diagnosis:

FORMAL DIAGNOSTIC TERMS FOR GENDER DYSPHORIA

Classification	Diagnostic term
International Classification of Diseases, 10th edition (ICD-10)	Transsexualism
Diagnostic and Statistical Manual of Mental Disorders, 4th edition (DSM-IV)	Gender identity disorder (GID)
DSM, 5th edition (DSM-5; American Psychiatric Association, 2013)	Gender dysphoria

- **Prevalence** (largest epidemiological review): male-to-female (transwomen) 1:11,900 to 1:45,000; female-to-male (transmen) 1:30,400 to 1:200,000 — **transwomen are more prevalent than transmen**.
- **Prevalence is rising** with public awareness and research, as is the number of **transgender and gender-nonconforming (TGN) youth** over the last decade.
- **World Professional Association for Transgender Health (WPATH)** treatment guidelines have been published since 1979; **Version 8 Standards of Care (SOC8)** appeared in 2022.
- **Gender-confirming surgery (GCS)** in transmasculine individuals may include mastectomy, hysterectomy, oophorectomy, metoidioplasty, phalloplasty and testicular or erectile prostheses; because genital and urinary tracts are intimately related, **genital surgery affects the urinary tract**.

UROLOGIC ISSUES IN GENDER-NONCONFORMING TRANSMASCULINE YOUTHS

- **Presentation has increased** over two decades, with a **bimodal peak**: children with severe GD, and adolescents at or just after puberty.
- **Atypical presentation** (mood and behavioural problems) is common, as is bullying and harassment at home, school and in the community.
- A matched retrospective cohort study showed a **substantially increased risk of depression, anxiety disorder, suicidal ideation, suicide attempt and self-harm**.
- They rarely present to (paediatric) urology, but **guidelines exist** for social transition, puberty blockers, gender-affirming hormones and surgery.
- **Age for surgery:** in most countries, the age of consent; in some, **chest surgery and gonadectomy from 16 years**. More adolescents are expected to present for gender-affirming surgery (GAS).
- **Urological problems are rare**. Some develop **lower urinary tract symptoms (LUTS) from altered toilet behaviour**: avoiding public toilets can lead to an **underactive detrusor** (unpublished observation).

- **Early hormonal care alters later surgery** (unpublished observations): the **oestrogen-naïve vagina makes vaginectomy much more complicated**, and **underdeveloped labia majora leave less tissue for scrotal reconstruction**.

Concept — *Treating youth early prevents female pubertal development, and the tissue that development would have produced is the tissue the reconstructive surgeon later needs; **early medical benefit is paid for in surgical difficulty**.*

UROLOGIC ISSUES IN TRANSMASCULINE INDIVIDUALS

- A long pathway of **staged invasive procedures**: secondary sex characteristics, removal of female and creation of male primary sex organs, secondary prosthetic surgery, and non-genital, non-breast procedures.
- **Care is patient-based, not provider-based**: the dichotomous view of gender is no longer central, and patients choose their own goals from a "**menu**" rather than receiving the provider's assumed ideal — so many requests lie between the binary poles.
- Most nonetheless seek four goals: **a normal-looking male genital appearance; voiding standing; preserved erotic sensation; penetrative sexual activity**.
- **A dedicated urologist in the gender team** is advisable.

Diagnosis and Preoperative Care

- **Preoperative urological screening is mandatory**, because GCS affects the lower urinary tract through vaginectomy, pelvic floor reconstruction and urethral lengthening.

Diagnosis

- GD is **diagnosed according to WPATH guidelines**, in a GD centre with a full **multidisciplinary team (MDT)**: psychologists, psychiatrists, endocrinologists, gynaecologists, social workers, plastic surgeons and urologists.
- GCS, if wished, is **often, but not necessarily, the last and most considered step**.

Gender-Confirming Surgery

- After confirmed diagnosis and **extensive counselling**, the patient may choose GCS.
- **Most transgender individuals never undergo external GCS** (a minority had surgically transitioned in a 2011 United States national survey).

Preoperative Care

- **Urethral lengthening, vaginectomy and pelvic floor reconstruction each impair postoperative voiding**; any pre-existing **lower urinary tract dysfunction (LUTD) must be diagnosed and treated before surgery**.
- **Work-up**: history + **uroflowmetry with post-void residual (PVR) + renal ultrasound**, excluding stones or hydronephrosis, which are harder to treat after GCS, especially after urethral lengthening.
- **WPATH: at least 12 months of continuous hormone therapy before masculinising GCS**. The main reason is that **testosterone induces clitoral hypertrophy**, with growth saturating at about **2 years**.
- **Hormone-blocking agents** — gonadotrophin-releasing hormone (GnRH) agonists such as triptorelin (Decapeptyl) — are increasingly used in under-age individuals, giving time to explore gender nonconformity (effects on surgery below).

HORMONAL TREATMENT AND GENITAL SURGERY: WHAT HELPS AND WHAT HINDERS

Agent	Helps surgery	Hinders surgery
Testosterone (topical or systemic)	Clitoral hypertrophy — more tissue for metoidioplasty and urethral lengthening	Vaginal atrophy — vaginectomy more difficult
GnRH agonist (puberty blockade)	Prevents irreversible female sex characteristics; facilitates transition	Prevents labia majora growth — less tissue for scrotoplasty

- **Informed consent** covers urological and general complications:

COMPLICATIONS TO INCLUDE IN CONSENT, BY PROCEDURE

Procedure	Principal complications
Urethral lengthening	Urethral fistula and stricture (most prominent)
Vaginectomy	Significant blood loss; postoperative mucocele. Rare but serious: injury to rectum, bladder, urinary sphincter
All of the above	Secondary LUTD and urinary tract infection (UTI)
Later stages	Implant-related complications

GENDER-CONFIRMING SURGERY IN TRANSMASCULINE INDIVIDUALS CAN CONSIST OF

1. **Removal of feminine anatomic structures:** mastectomy, oophorectomy, hysterectomy and vaginectomy.
2. **Creation of masculine primary sex organs (genitoperineal transformation):** phalloplasty with or without urethroplasty, metoidioplasty with or without urethroplasty, scrotoplasty, monsoplasty and perineal reconstruction.
3. **Implantation** of testicular and erectile prostheses.

Concept — Every step of genitoperineal reconstruction can worsen voiding, so **the bladder is assessed and treated first**, and testosterone is given long enough to grow the clitoris even though it simultaneously makes the vagina harder to remove.

Removal of Female Anatomic Structures**Mastectomy, Hysterectomy, and Ovariectomy**

- **Mastectomy is usually first: subcutaneous** — glandular tissue removed, overlying skin and part of the nipple preserved; type by anatomy and surgeon.
- **Ovariectomy and hysterectomy preferably laparoscopic** (less morbidity and scarring); otherwise, or on conversion, **Pfannenstiel incision**.
- **Ureteric injury:** the ureters cross the iliac vessels and run to the trigone **beneath the uterine artery**. Injury → the urologist is called for **immediate repair**, the type depending on extent and location.
- **Bladder injury** (small risk) → **immediate repair**.
- **Remove the entire cervix and, if possible, the proximal third of the vagina** at hysterectomy, to facilitate later vaginectomy from the perineum.
- The three may be **combined in one procedure** to minimise operations.

Vaginectomy

- Usually **combined with genitoperineal transformation**, not performed separately.

- **A hazardous procedure.** Major complications, in descending frequency: **haemorrhage requiring reintervention > bladder injury > rectal injury.**
- Given this severity, and because **vaginectomy is not necessary**, it is discussed; some keep the vagina for intimacy.
- **No vaginectomy → no urethroplasty:** lengthening the urethra without vaginectomy significantly increases the risk of **urethrovaginal fistula.**
- **Two options:** total vaginectomy; or **leave the upper vagina and fulgurate its mucosa** to prevent a mucocele.

Technique of total vaginectomy.

1. **Hydrodissection:** the plane between the anterior vaginal wall and the urethra–bladder is infiltrated with **dilute xylocaine (lidocaine)** (Fig. 128.1).
2. Introitus **incised circumferentially** → vagina grasped with two mosquito clamps anteriorly and two posteriorly.
3. **Rectovaginal plane:** usually easy, by **blunt digital dissection** → a gauze is placed to **protect the rectum** (Fig. 128.2).
4. **Urethrovaginal plane:** **sharp dissection with tenotomy scissors; no diathermy**, to avoid thermal injury to the sphincter (Fig. 128.3A) → then the bladder–vaginal plane by blunt and sharp dissection. A **submucosal plane in the vaginal wall protects the bladder**; a bladder injury is usually **midline and closed primarily**; if the integrity of the ureteric orifices is in doubt → **cystoscopy**.
5. **Lateral walls:** mosquitoes replaced by two **Jean-Louis Faure clamps** (Fig. 128.3B) → levator dissected off the lateral wall → **high-energy cautery for haemostasis of the paravaginal plexus**; persistent bleeding → Vicryl (polyglactin 910) 3-0 sutures.
6. Rectal gauze removed → haemostasis; **haemostatic agents in the space between rectum and bladder.**
7. **Levators approximated** in the midline with two to three Vicryl 1-0 sutures (Fig. 128.4) → **prevents later prolapse of bladder or rectum.**
8. **Bulbospongiosus closed over the fixed part of the urethra** → perineal wound closed in layers.

The best time to perform a vaginectomy remains unclear.

TIMING OF VAGINECTOMY: THREE STRATEGIES

Strategy	What is combined	Evidence and rationale
All-in-one	Mastectomy, ovariectomy, hysterectomy and vaginectomy with genitoperineal transformation	With phalloplasty: more major complications and transfusion than two-stage. With metoidioplasty: no excess
Vaginectomy early	Vaginectomy at hysterectomy–ovariectomy, or with temporary metoidioplasty; phalloplasty later	Vaginectomy blood loss and hypotension spare flap perfusion — theoretical; preliminary data show no fewer overall complications when metoidioplasty precedes phalloplasty
Two-stage (authors' preference)	Stage 1: mastectomy, ovariectomy, hysterectomy. Stage 2: vaginectomy with genitoperineal transformation	Authors' preferred technique

- The all-in-one excess is **confounded by reconstruction type**: seen with phalloplasty, not metoidioplasty.

Concept — *Vaginectomy is **the most dangerous and the only optional step**, yet it decides whether urethral lengthening is safe; its timing is chosen to keep its blood loss away from a free flap that depends on perfusion.*

Genitoperineal Transformation in Phalloplasty

Urethroplasty

- The transmasculine urethra has **three parts** (seen on retrograde urethrography, Fig. 128.11):

THE THREE PARTS OF THE URETHRA AFTER PHALLOPLASTY

Part	Extent	Construction	Male equivalent
Native urethra	Bladder neck → original external meatus	Original female urethra	—
Fixed part (pars fixa, perineal)	Original meatus → under the enlarged clitoris	Vestibular mucosa (ventral clitoris, meatus to glans) + inner labia minora → 3-cm plate tubularised over 18-French (Fr) silicone tube , running 3-0 polyglactin (Fig. 128.5)	Bulbar urethra , but without spongiosum
Phallic part (pendulous)	Within the phallus → neomeatus	Tube-in-a-tube : 3-cm flap skin island round a catheter, running 3-0 polyglactin (radial forearm free flap, Fig. 128.8); or a separate pedicled or free flap	Penile urethra , but without spongiosum

- **Anterior vaginal mucosa flap** (only at vaginectomy): avoids a **ventral suture line at the native–fixed anastomosis**, reducing fistula. Against: a **random flap** without solid pedicle → **flap necrosis**; authors suspect this hypertensile tissue causes **sacculation and postvoid dribbling** and do not use it (unconfirmed).

Sequence after construction of the fixed part.

1. **Clitoral glans mucosa removed**, aided by hydrodissection with dilute xylocaine and adrenaline.
2. **One dorsal clitoral nerve dissected for 2–3 cm** — found at the **1 and 11 o'clock** positions; side chosen = **side of the phalloplasty vascular anastomosis**.
3. **Prepubic incision** at the future phalloplasty site → **clitoris fixed to the pubic symphysis** with Vicryl 2-0 (Fig. 128.6).
4. After the vascular anastomosis, the **clitoral nerve is coapted to a cutaneous nerve of the flap**.

5. Distal end of the fixed part prepared for the phallic anastomosis with **separate polyglactin 4-0 sutures**.
6. **Bulbospongiosus muscles sutured in the midline over the fixed urethra** (Fig. 128.7).
7. Phallic urethra: proximal end joined to the fixed part using the pre-placed interrupted 4-0 sutures → distal end sutured to phallic skin as the **neomeatus**.
 - **Bulbospongiosus cover** gives **vascular support** (fewer fistulae) and **expels the last drops** (less dribbling).

Concept — *The neourethra is **skin or mucosa lying in fat, without corpus spongiosum**; wrapping bulbospongiosus over the fixed part is a deliberate attempt to restore the vascular cover and the expulsive squeeze that spongiosum and a native bulbospongiosus would normally provide.*

Scrotoplasty

- **Hair-bearing labia majora form the neoscrotum.**
 - **Simple midline closure** over the former introitus leaves an **unaesthetic scrotum between the legs**; the cisgender scrotum lies **in front of the legs** — hence flap rotation (Ghent-type scrotoplasty, Fig. 128.9).
1. Labia majora **incised dorsally and semicircumferentially** and mobilised → bilateral labial flaps supplied by **branches of the external pudendal artery**.
 2. Remnants of non-hair-bearing labia minora and clitoral foreskin resected from the flaps.
 3. Flaps **turned 180 degrees medially** and sutured at the top of the perineal skin.
 4. Perineum closed with subcutaneous **polyglactin 1-0** — **large calibre because tension is significant** → flaps sutured to each other and perineal skin closed with absorbable Vicryl 3-0.

Phalloplasty

- **Many free and pedicled flaps have been described** — meaning no single flap achieves all the ideal goals.
- Usually by a **plastic surgeon**; flap choice by expertise, patient preference and local anatomy (previous pelvic or genitoperineal surgery, **body mass index (BMI)**).

Metoidioplasty

- An **alternative to phalloplasty for transmasculine individuals for whom penetration is a secondary goal or not a goal**.
 - **First described by Lebovic and Laub (1999)** and popularised by Perovic at about the same time.
1. **Hormonally enlarged clitoris** → **chordee divided**.
 2. Clitoris released from its **labia minora attachments**.
 3. **Wide ventral urethral plate released** — the plate is what bends the clitoris downward → maximal length gained, but the **urethra must now be lengthened**.
 4. Fixed part reconstructed as described above.
 5. Urethra extended to the tip of the glans with a **transverse preputial island flap or a labia minora flap**, which may be combined with **buccal mucosa** (authors: transverse preputial island flap).
- Variants differ in the **extent of clitoral release** and in scrotal reconstruction.

METOIDIOPLASTY VERSUS PHALLOPLASTY

	Metoidioplasty	Phalloplasty
Phallus	Patient's own clitoris; 3–8 cm micropenis , realistic appearance	Flap-built phallus
Erection	Spontaneous (engorgement)	Requires erectile prosthesis
Erotic sensation	Largely maintained	Depends on clitoral–flap nerve coaptation
Standing voiding	Most often achieved	A principal goal
Penetration	Usually not possible	A principal goal
Scars	No donor-site scar	Visible donor-site scars — a reason some choose metoidioplasty

- **Adjunctive enlargement** (topical testosterone ± clitoral vacuum or traction) is debated: since results depend on clitoral hypertrophy, extra enlargement may help, but **no head-to-head long-term comparison exists**; the authors do not use it, judging it of limited value after their **recommended minimum of 2 years of systemic testosterone**.
- **Complications** (limited literature; minor more frequent than major), by **Clavien–Dindo**:

COMPLICATIONS OF METOIDIOPLASTY BY CLAVIEN–DINDO GRADE

Grade	Examples	Management
Minor (I–II)	Haematoma, infection, delayed wound healing	Usually conservative
Major (III–V)	Permanent urethral fistula, urethral stenosis, flap necrosis	Surgical revision

Concept — *Metoidioplasty trades length for native function: building on the patient's own erectile, innervated clitoris preserves spontaneous erection and sensation but can never provide penetrative size.*

Postoperative Care, Follow-up, and Urethral Complications**Early Postoperative Care**

- Focus is **recovery of micturition**. After urethral lengthening:
 1. **Suprapubic catheter (SPC) + trans-neourethral catheter** left in situ.
 2. **Neourethral catheter removed at 14–21 days** → patient voids with the SPC in place.
 3. SPC closed with a tap, **opened after every void to measure PVR**, recorded in a bladder diary.
 4. Voiding may be delayed by **swelling or neurogenic bladder impairment after vaginectomy** → SPC kept until normal function returns. Other care is standard urogenital care.

Transurethral Catheterization in Transmasculine Individuals After Gender-Confirming Surgery

- **Why difficult**: an **abrupt curve** into the native urethra, and **two small sinuses toward the Bartholin's gland orifices** on each side of the native meatus, in which the catheter tip catches.
- Elective catheterisation **by the urologist**:
 1. **One blind attempt with a Coudé catheter** — abort at any difficulty.
 2. Second attempt with a **mandrin (stylet)**, which negotiates the sharp curve at the native meatus.
 3. Any difficulty → **stop** (to avoid urethral trauma and later stricture) → **flexible cystoscopy with guidewire** into the bladder → **open-tip catheter over the wire**.
- **Emergency**: a **suprapubic catheter is always a good solution**.

Urethral Complications in Transmasculine Individuals After Gender-Confirming Surgery

- **Urethral complications are frequent after phalloplasty**, affecting a large proportion of patients.

Urethral fistula.

- A **common early complication**, related to **delayed wound healing of the phallus**.
- **Site: base of the phallus (commonest) > shaft**; **perineal fistulae are rare** because local tissue covers the urethra well there. Without vaginectomy, **urethrovaginal fistulae** can arise (shown on voiding cystography, Fig. 128.10).
- 1. Early: the catheter is **visible through a wound dehiscence**.
- 2. **Maintain urinary diversion** (urethral or suprapubic) to avoid further wound maceration until the wound heals.
- 3. Diversion removed → voiding through the fistula; **small fistulae often close spontaneously**.
- 4. **Persisting beyond 3 months = permanent** → surgical closure.
- **Repair principles**: excise the tract, close urethra and skin, and **interpose a healthy layer** — **no dartos exists, so only local fat** is available. **Catheter for 10–12 days after repair**.
- Healing is slower than in males: **flap skin**, not **mucosa within spongiosum**.

Urethral stricture — incidence.

- **Common**, again from **phallic healing problems**: most lie in the **phallic urethra, at the anastomosis or at the meatus** — the latter two at the **flap extremities, where blood supply is poorest**.
- **Commonest site: the anastomosis between the fixed and phallic parts** — a **mucocutaneous junction** prone to stricture from delayed healing.
- Most anastomotic strictures are **small and appear early** after phalloplasty; symptoms and signs are as in males.

Diagnosis.

- **Plateau uroflow curve** → **retrograde urethrography (RUG) is necessary**: confirms, locates and measures; three urethral parts distinguishable (Fig. 128.11) → **urethroscopy** if doubt.

Management — principles.

- **Defer any treatment ≥6 months after phalloplasty**, until wound healing stabilises; reconstruction during acute inflammation is useless.
- **Severe obstruction or retention** → **SPC first line**.
- **Perineal urethrostomy** if a long wait is expected, SPC causes severe complaints (e.g. bladder spasm) or is refused; closed at or after successful reconstruction.
- Choice depends on **previous interventions, stricture location, length and concomitant phallic problems** (e.g. flap contracture). The same techniques apply **after metoidioplasty**.

TREATMENT OPTIONS FOR URETHRAL STRICTURE AFTER PHALLOPLASTY OR METOIDIOPLASTY

Option	Indication	Key technical points and verdict
Intermittent dilation	Low-grade meatal stenosis	Interval set by the interval of recurrence. Repetitive dilation for anastomotic strictures is not curative — discourage
Direct vision internal urethrotomy (DVIU)	Endoscopic first option	Sedation (phallus not yet sensate) or spinal/general anaesthesia → guidewire (avoids false passage) → cold knife at 4, 8, 12 o'clock to fat until resectoscope passes → 18-Fr catheter 2 weeks (skin must heal). Low success; repeats useless; earlier after phalloplasty = lower success

Option	Indication	Key technical points and verdict
Urethral reconstruction (all options below)	Short (<3 cm) strictures refractory to dilation or DVIU , and longer strictures at any segment	Recurrence higher than urethroplasty in native males
Meatotomy	Short meatal stricture where dilation impossible or unwanted	Ventral incision to healthy urethra; phallic to urethral skin → slightly hypospadiac meatus
Heineke-Mikulicz urethroplasty	Short, non-obliterative strictures, any segment	Longitudinal incision, transverse closure
Excision and primary anastomosis (EPA)	Short, nearly obliterative strictures at the fixed-phallic anastomosis	Penoscrotal-angle incision → meatal sound marks distal end → ventral opening to healthy fixed urethra (complete obstruction: sound via perineal urethrostomy or suprapubic tract marks proximal end) → resect, spatulate, mobilise, tension-free anastomosis
Free graft urethroplasty	Anastomotic site; fixed part; especially after metoidioplasty	Fat cannot support a graft → at the anastomosis, reinforce with a local fasciocutaneous flap ; at the fixed part, dorsal inlay quilted onto the clitoral corpus
Two-stage urethroplasty (standard)	Longer phallic strictures ; complex anastomotic strictures; complex post-metoidioplasty strictures	Stage 1: open and marsupialise ; narrow or scarred plate replaced or augmented with full-thickness skin graft or buccal mucosa . Stage 2, ≥6 months later if healed: retubularise over a catheter
Flap urethroplasty	Complex or recurrent phallic stricture → complete substitution	Local phallic or scrotal flaps limited (little mobile tissue; deformity) → radial forearm free flap or superficial circumflex iliac artery (SCIA) free or pedicled groin flap ; phallic problems (e.g. graft contracture) corrected concurrently
Definitive perineal urethrostomy	Refractory stricture, no wish for further reconstruction	Midline perineal incision → fixed part opened 4–5 cm → edges to perineal skin (Fig. 128.12)

- **What the table tests:** matching technique to site, length and obliteration. **Pattern:** the urethral bed is **fat, not spongiosum**, so endoscopic and dilatational treatments fail and grafts need a flap for blood supply.
- **Trap:** repeating dilation or DVIU for an anastomotic stricture, or treating before 6 months.

Urethral hair growth.

- A skin-flap urethra (the usual case) can grow hair; usually harmless, but complications are **important and difficult** — **so prevent**.
- **Prevention:** **non- or minimally hair-bearing flap** (e.g. **ulnar forearm** for a radial forearm flap); otherwise **preoperative laser depilation** (strongly recommended).
- Substantial growth → **hair bezoar with stone formation** → difficult voiding, local irritation, urethral discharge. **Bezoar and calculus formation, not hair growth itself, causes the voiding complaints.**

1. Phallic urethra **opened ventrally**, calculi removed.
2. Urethral edges sutured to the opened phallic skin.
3. Urethral hair treated by **laser epilation**.
4. Later: plate **retubularised**, as in the second stage of a two-stage urethroplasty.

Postvoid dribbling.

- **Nearly universal.** The neourethra is a **rigid tube without spongiosum elasticity**, and reconstructed bulbospongiosus expels the last drops poorly.
- **Advice:** **perineal pressure plus squeezing of the phallus** at the end of micturition.

- **Severe dribbling** → urethrography or urethroscopy to exclude **urethral sacculation or a pseudodiverticulum from a vaginal remnant**; resection may resolve or reduce it.

Concept — *Every urethral complication traces back to the flap: skin in fat, with poorly perfused ends and no spongiosum, explains why fistulae and strictures cluster at the flap extremities, heal slowly, need longer catheterisation, resist endoscopic cure, and why dribbling is almost universal.*

Testicular and Erectile Prosthetics

- Implanted as a **separate procedure**; testicular prostheses are **usually implanted with the erectile prosthesis** when both are wanted.

TESTICULAR VERSUS ERECTILE PROSTHESIS

	Testicular prosthesis	Erectile prosthesis
Earliest timing	6 months after scrotoplasty , when the scrotum has healed	1 year after phalloplasty , provided the phallus has protective sensation and there are no urethral problems
Size or number	Scrotum usually accepts 17-mL prostheses	One cylinder usually sufficient for axial rigidity; two cylinders' bulk risks skin and urethral ischaemia, erosion and perforation
Technique	Small incision at scrotal base → blunt pouch → fixation to prevent displacement	No tunica albuginea to anchor the cylinder → rear tip extender or Dacron vascular prosthesis at the cylinder base, fixed to the pubic periosteum just lateral to the urethra

- **Malleable versus inflatable**: both possible; **malleable is technically easier with a lower risk of technical failure**.
- **Infection and/or erosion are relatively frequent** (hypovascular phallus, poor cover) → **urgent explantation**; reimplant **≥6 months later**.
- **Technical malfunction and malposition** are other frequent reasons for revision.
- **Dedicated prostheses (since 2016; Zephyr Surgical Implants [ZSI], Switzerland)**: one larger cylinder, **silicone glans cap**, **titanium base plate sutured to the pubic bone** (Fig. 128.13).

DEDICATED PHALLOPLASTY PROSTHESES

Device	Type	Experience
ZSI 475 FTM (female-to-male)	Three-piece inflatable	Complication rates remain high: substantial explantation within the first 18 months (protrusion, infection, malposition, mechanical failure)
ZSI 100 FTM	Malleable single piece	No series; authors see more protrusion and pain

- Authors use them for **all primary placements**: sturdier fixation, less release from the pubis (final result, Fig. 128.14).

Concept — *A neophallus lacks what a penis gives a prosthesis — a tunica to anchor it and a robust, well-perfused cover to protect it — so implants are single, anchored to bone, delayed until sensation returns, and still carry high infection, erosion and explantation rates.*

CONCLUSION

- **GCS is done in few specialised centres, but many have undergone it**, so any urologist may meet them.
- Every urologist should be able to **manage the acute urological problem**; thereafter, **refer complications to an experienced centre**.